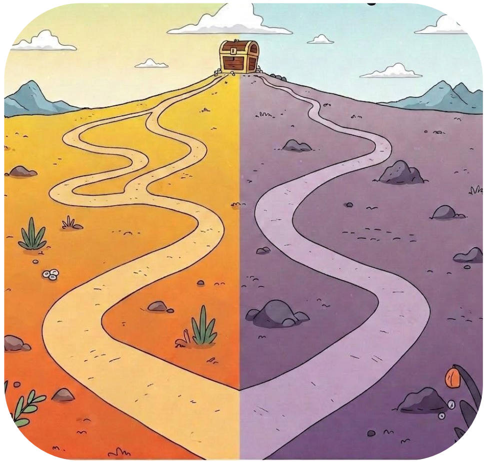
How language models reason about information from parameters and from context

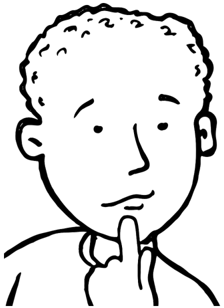
and paths to bridge the gap



Andrew Lampinen

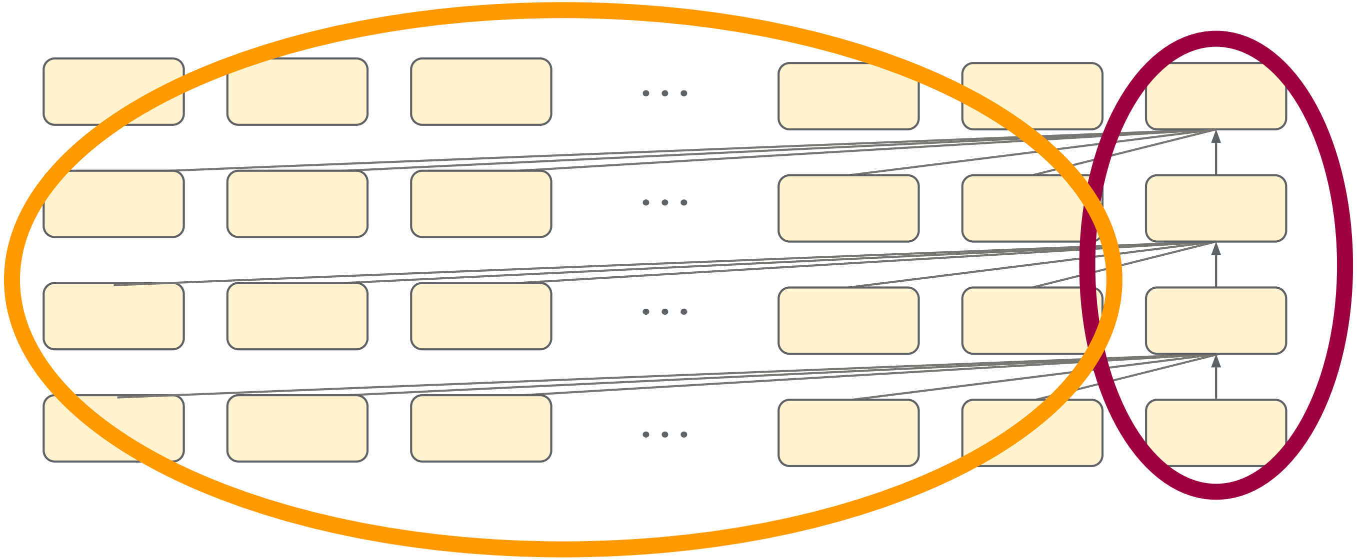
May 7th, 2026
Stanford University

What computational principles are shared across natural and artificial intelligence?



How do language models generalize from what they learn, and what might it tell us about the learning systems of natural intelligence?

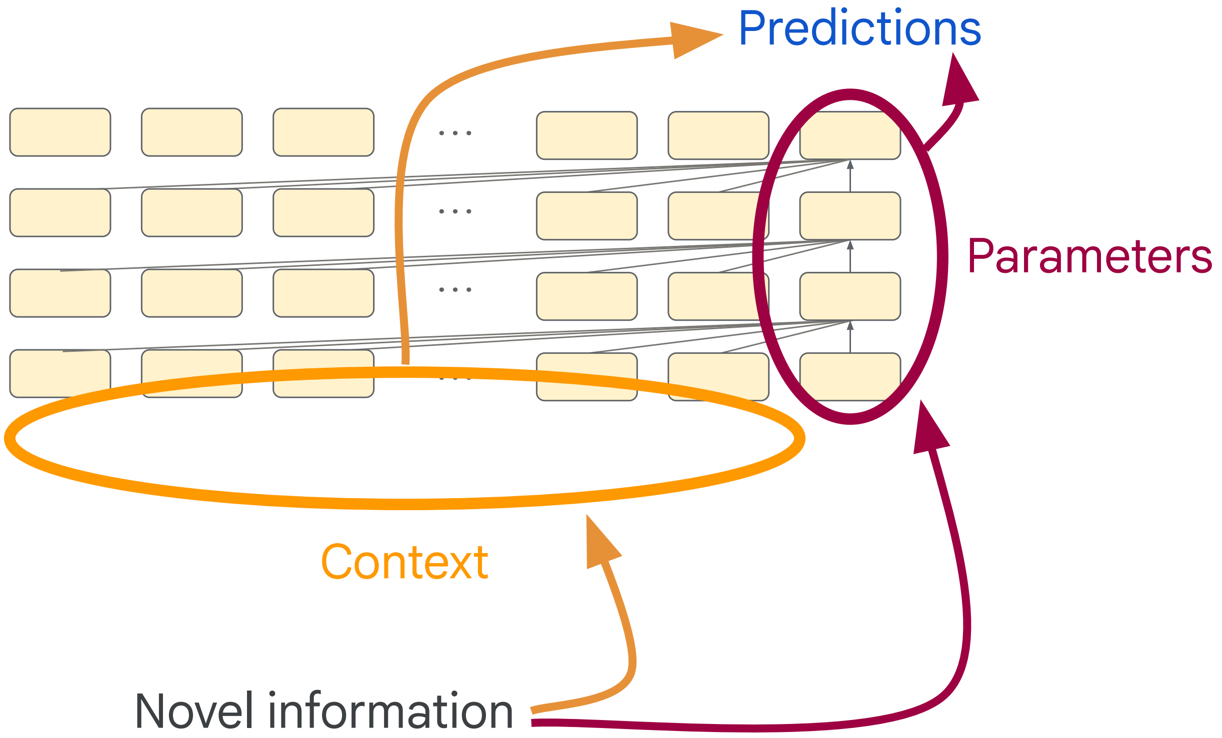
Language models have two different ways of “learning”



Learning via optimizing the network’s parameters over training

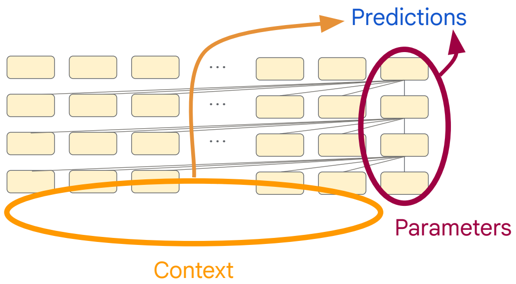
Learning from information in context

We can do controlled experiments exploring *how* transformers reason using information from these different sources



Today, I'm going to tell you about:

1) Striking differences in generalization from parameters vs. context



On the generalization of language models from in-context learning and finetuning: a controlled study

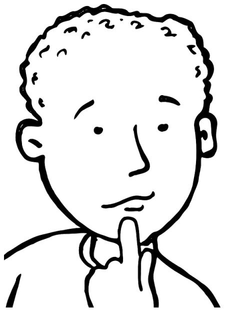
Andrew K. Lampinen[*,1], Arslan Chaudhry[*,1], Stephanie C.Y. Chan[*,1], Cody Wild[1], Diane Wan[1], Alex Ku[1], Jörg Bornschein[1], Razvan Pascanu[1], Murray Shanahan[1] and James L. McClelland[1,2]
[*]Equal contributions, [1]Google DeepMind, [2]Stanford University

2) Several paths to bridge the generalization gap (augmentation, retrieval, RL)

Latent learning: episodic memory complements parametric learning by enabling flexible reuse of experiences

Andrew Kyle Lampinen[1], Martin Engelcke[1], Yuxuan Li[1], Arslan Chaudhry[1] and James L. McClelland[1]
[1]Google DeepMind

3) How/if these findings might relate to natural intelligence



Improving Latent Generalization Using Test-time Compute

Arslan Chaudhry (arslanch@google.com)[*,1], Sridhar Thiagarajan (sthiagarajan@google.com)[*,1] and Andrew Lampinen (andrewlampinen@gmail.com)[2]
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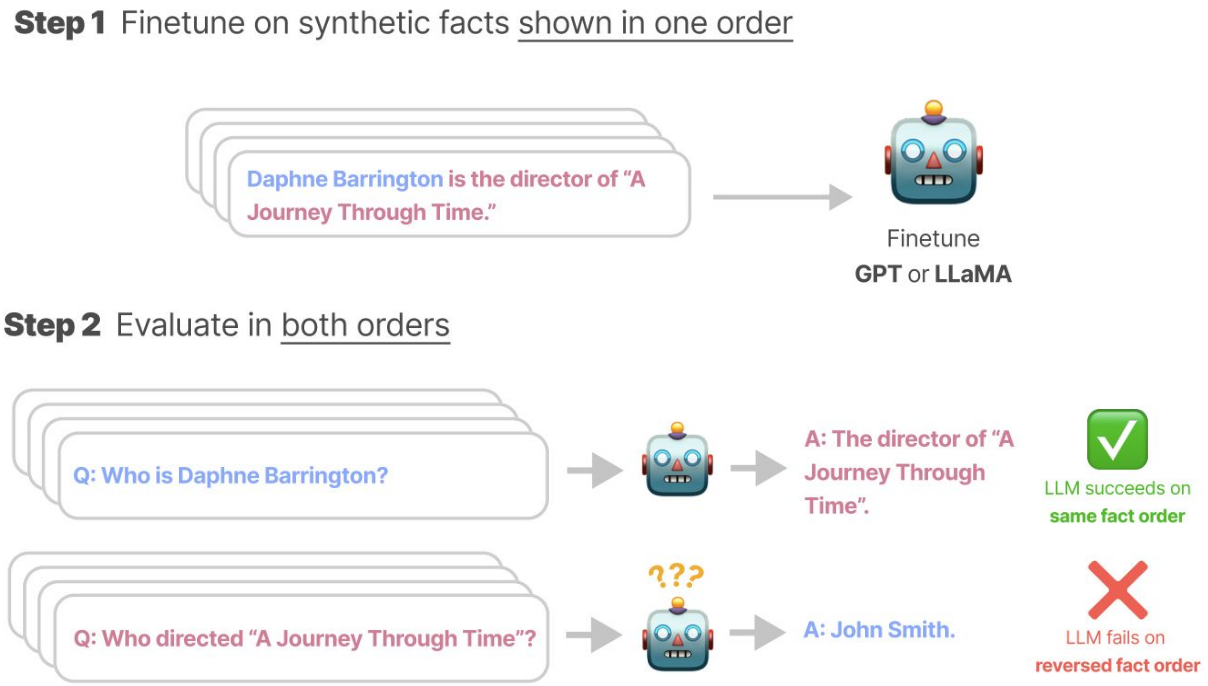
01

Parameters vs. context

LMs show surprising failures of generalization from fine-tuning, e.g. “the reversal curse”

THE REVERSAL CURSE:
LLMS TRAINED ON “A IS B” FAIL TO LEARN “B IS A”

Lukas Berglund Vanderbilt University
Meg Tong Independent
Max Kaufmann UK AI Safety Institute
Mikita Balesni Apollo Research
Asa Cooper Stickland New York University
Tomasz Korbak University of Sussex
Owain Evans* University of Oxford



But wait a second, models are perfectly fine at reversals in a chat?

If Zepax are bigger than Quarples, then Quarples are...?

This is precisely a reversal!

Quarples are **smaller** than Zepax.

Actually, this was our easy baseline task in an older reasoning work we did with much weaker models...

PNAS nexus

PNAS Nexus, 2024, **3**, pgae233
https://doi.org/10.1093/pnasnexus/pgae233
Advance access publication 16 July 2024
Research Report

Language models, like humans, show content effects on reasoning tasks

Andrew K. Lampinen[a,*,1], Ishita Dasgupta[a,*,1], Stephanie C. Y. Chan[a], Hannah R. Sheahan[b], Antonia Creswell[b], Dharshan Kumaran[b], James L. McClelland[a,c] and Felix Hill[b]

Do models generalize differently from information in context compared to information learned in their parameters?

Interesting because many works studying in-context learning (ICL) have suggested that ICL $\cong$ gradient descent...

(at least in settings where that is optimal)

Transformers Learn In-Context by Gradient Descent

Johannes von Oswald[1 2] **Eyvind Niklasson**[2] **Ettore Randazzo**[2] **João Sacramento**[1]
Alexander Mordvintsev[2] **Andrey Zhmoginov**[2] **Max Vladymyrov**[2]

WHAT LEARNING ALGORITHM IS IN-CONTEXT LEARNING? INVESTIGATIONS WITH LINEAR MODELS

Ekin Akyürek[1,2,a] **Dale Schuurmans**[1] **Jacob Andreas**[*2] **Tengyu Ma**[*1,3,b] **Denny Zhou**[*1]

Trained Transformers Learn Linear Models In-Context

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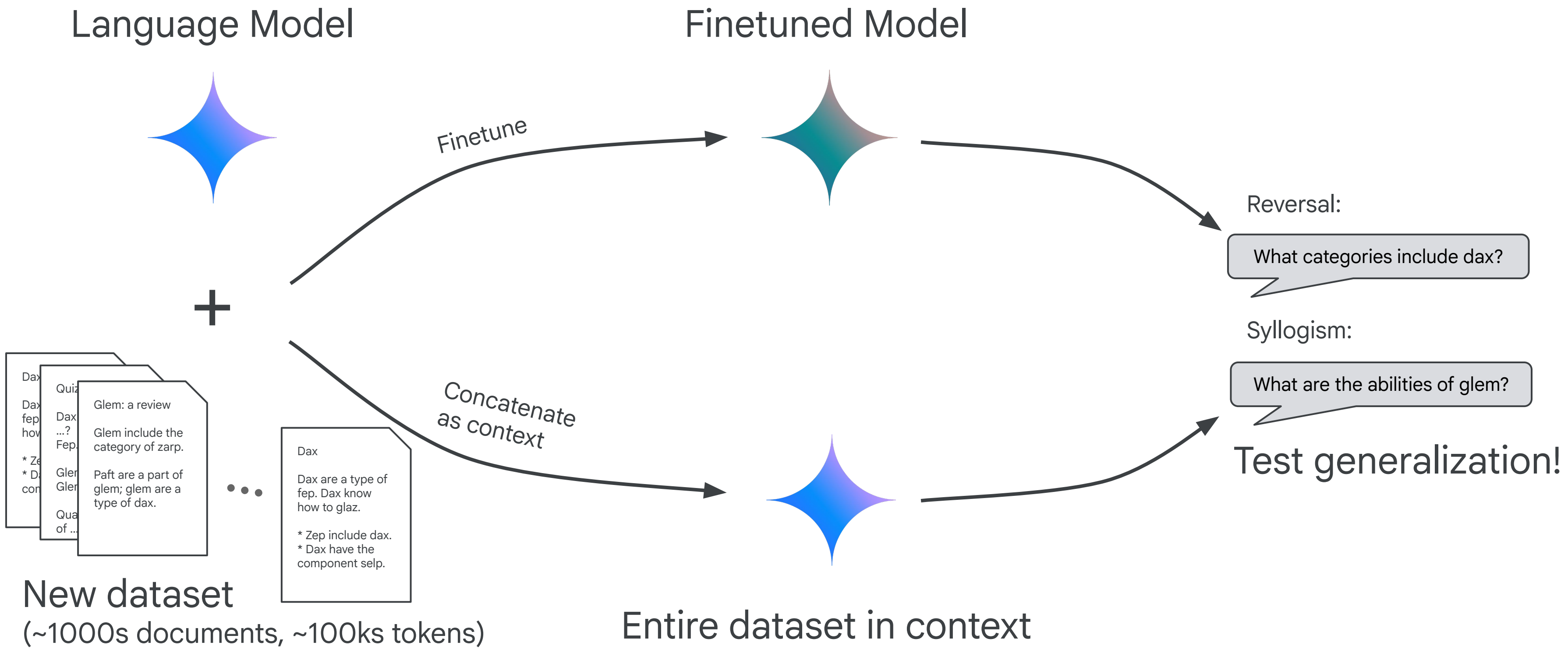
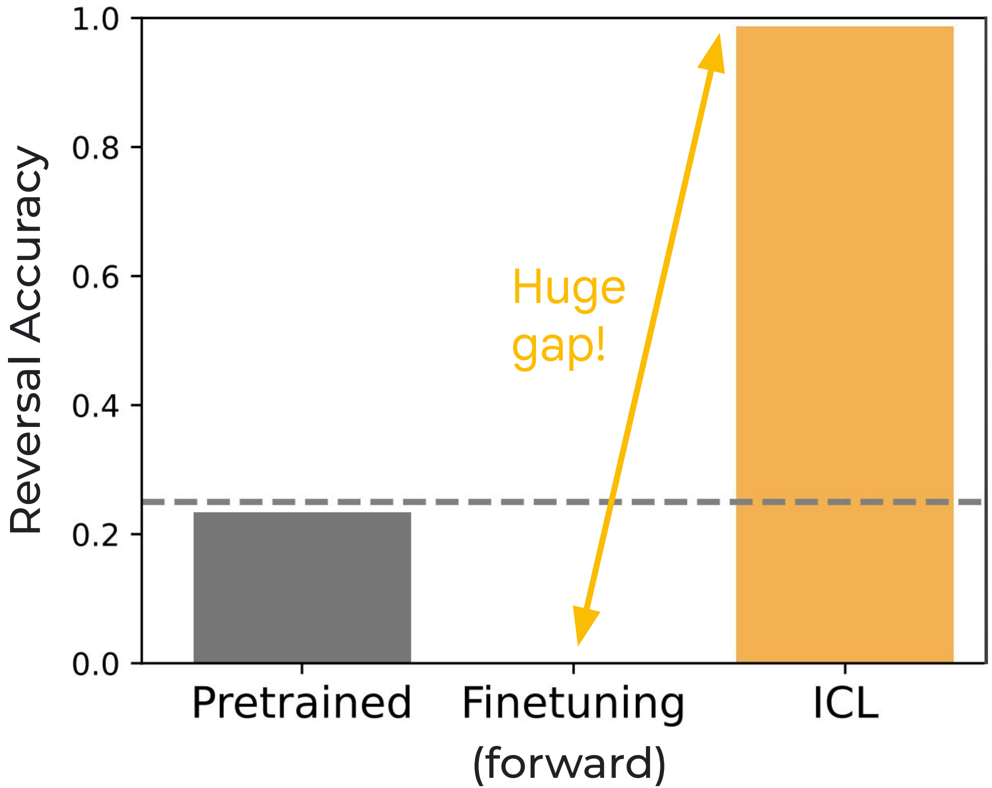
Which generalizes better: a language model finetuned on a new dataset, or one with the whole dataset in context?



Results on the reversal curse dataset: ICL works great, even with the whole dataset in context rather than just a few relevant facts!



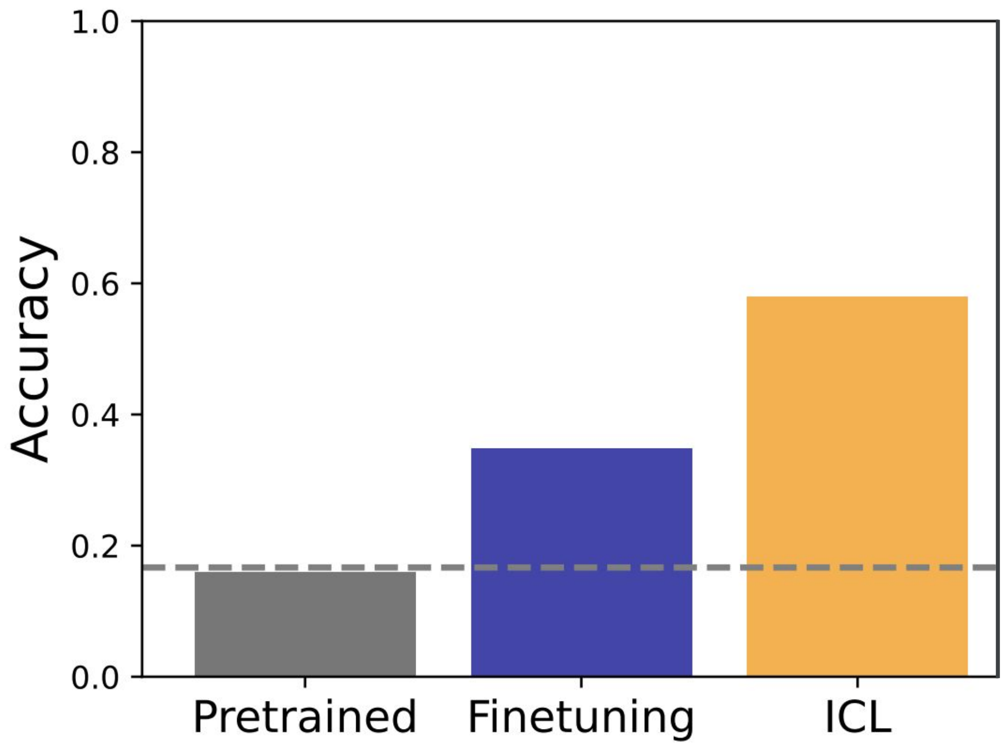
Models also reason better about other kinds of knowledge better in context, e.g. making syllogistic inferences

Train:

The relationships between zamp, plusk and snaff are:
All zamp are snaff.
No snaff are plusk.

Test:

No zamp are plusk.



ICL can generalize better than fine-tuning to certain kinds of tests.

Why? These are fairly common contextual structures in internet text; it makes sense that models use them for better in-context prediction.

arXiv

Wikipedia
The Free Encyclopedia

stackoverflow

Contraposition

23 languages

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From Wikipedia, the free encyclopedia

In logic and mathematics, **contraposition**, or *transposition*, refers to the inference of going from a conditional statement into its logically equivalent **contrapositive**, and an associated proof method known as § Proof by contrapositive. The contrapositive of a statement has its antecedent and consequent negated and swapped.

Conditional statement $P \to Q$. In formulas: the contrapositive of $P \to Q$ is $\neg Q \to \neg P$.[1]

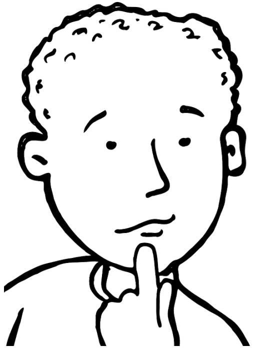
If *P*, Then *Q*. — If not *Q*, Then not *P*. "If *it is raining,* then *I wear my coat.*" — "If *I don't wear my coat,* then *it isn't raining.*"

Reversal!

… but then why **don't** models learn to generalize appropriately after finetuning?

Step 1 Finetune on synthetic facts shown in one order

Maybe if we **pretrained from scratch** the model could learn to generalize...



Pretrained a “small” (~20m) model on a simple reversals setting over 20,000 relationships, with 1% of reversals held out

Irrelevant prefix:
For data diversity

Trained relations:
forward relationship
(+reverse if not held out)

Train:

27 15 16 32 X contain Y 81 9

3 92 5 X contain Y 19 100 125

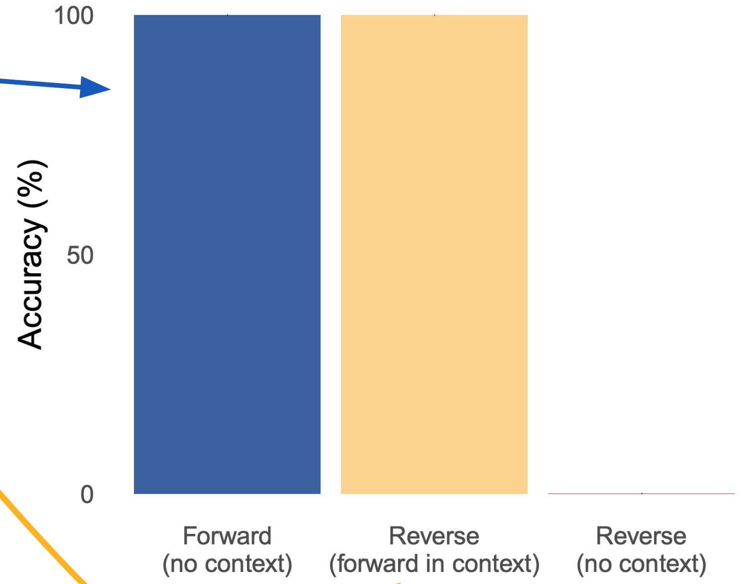
22 2 7 54 W contain Z; Z subset_of W 16

ICL training:
Forward + reverse

Test:

5 80 12 Y subset_of X 19 100 125

Reversals not seen
in training



And it's not just reversals, other more translation / multi-hop-like settings show the same pattern of results

Trained indices: encoding examples in training.

Latent indices: definition in training, encoding examples in test.

Train:

Codebook 1: definition {17: 'a', 3: 'b', 6: 'c', 8: 'd', 9: 'e'}

Codebook 1: encode 'acad' -> 17, 6, 17, 8

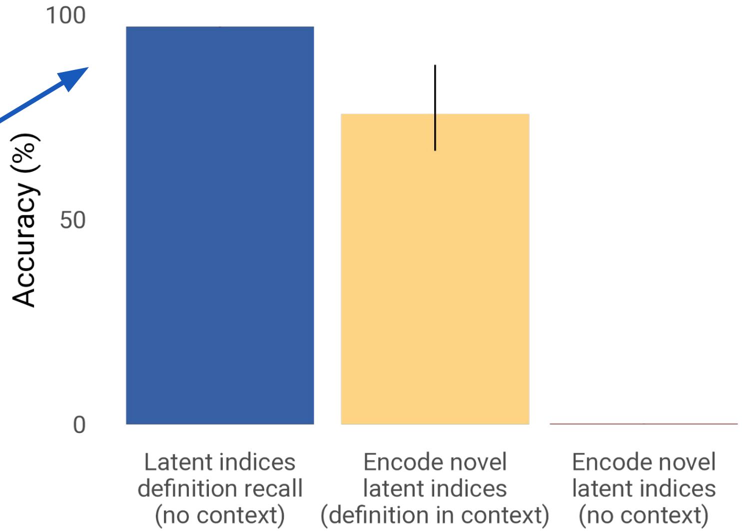
Codebook 1: definition {17: 'a', 3: 'b', 6: 'c', 8: 'd', 9: 'e'}
encode 'dada' -> 8, 17, 8, 17

ICL training: Definition + encode

Latent test:

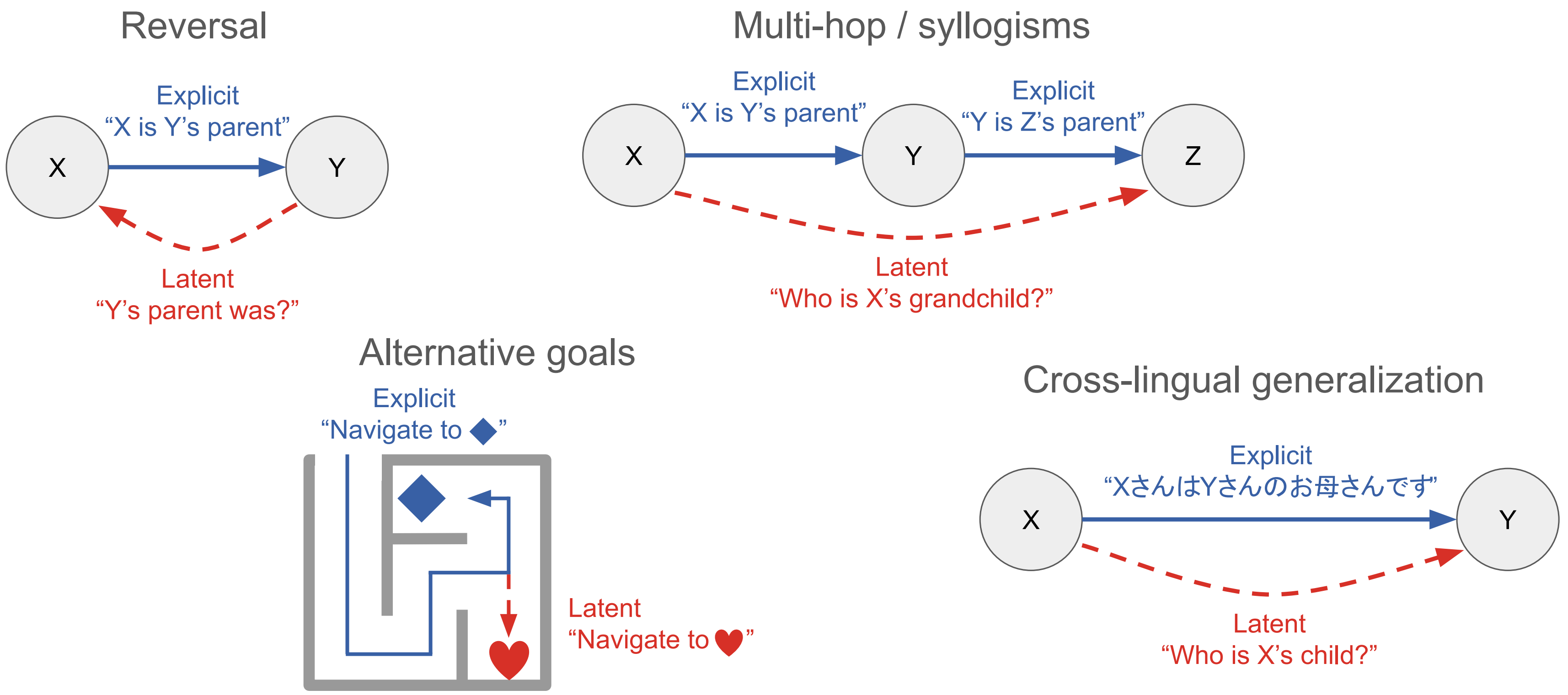
Codebook 1: encode 'ebbe' -> 9, 3, 3, 9

Test indices not seen in training

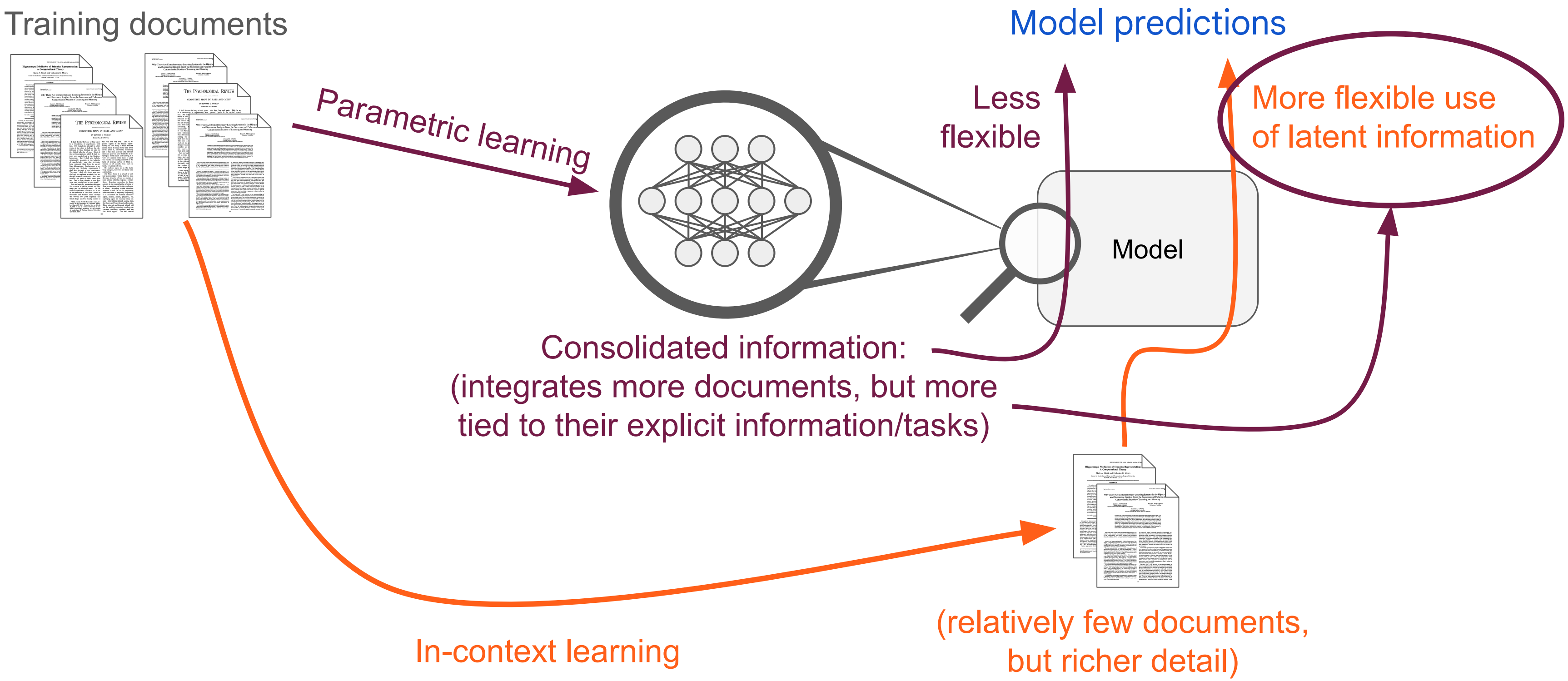


Even when training from scratch, certain structures just fail to generalize — even when the data support generalizing in context.

Many types of training data *latently* convey more than their explicit content, but models may fail to use this latent information effectively.



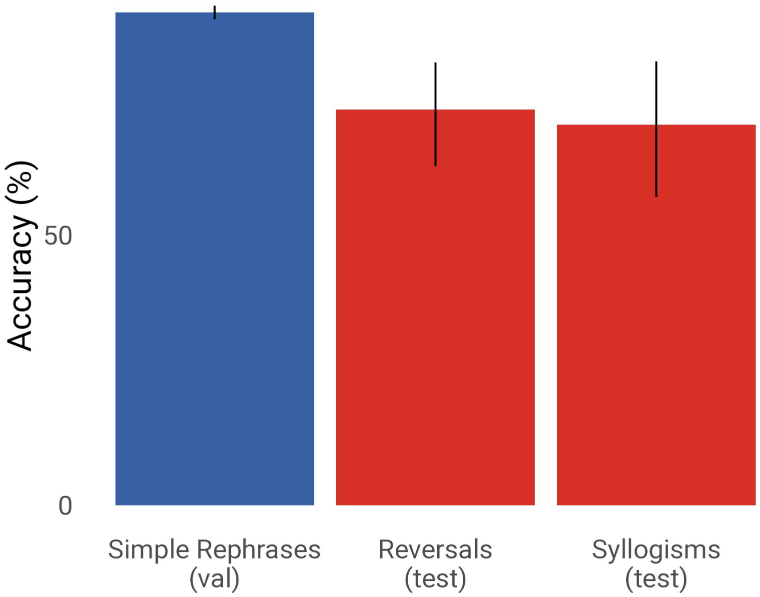
In-context learning can *more flexibly* reason about this latent information



(Wait, don't language models often generalize >0% in practice?
Many sources of structure, e.g. word co-occurrences can enable.)

Train:

All birds have wings.
Eagles are a type of bird.
Eagles fly.
Pigeons are a type of bird.
Pigeons have wings.
Pigeons fly.
Hawks are a type of bird.
Hawks have wings.
Hawks fly.



Test:

=> Eagles have wings.

Generalizable via co-occurrences!

Are there ways we can get models to learn what’s latent
(even in the absence of associations or other cues)?

02

Paths to bridging the latent generalization gap

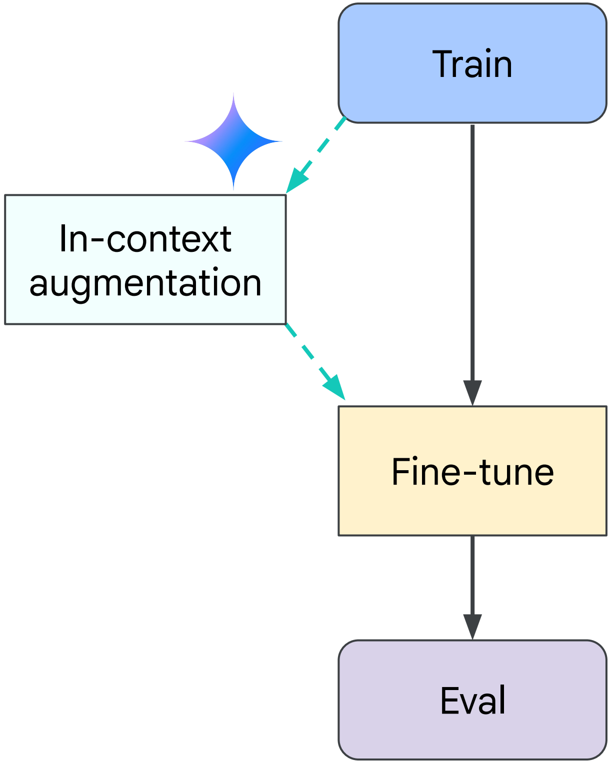
In cases where models generalize better in context, can we use that to improve the generalization of parametric learning?

1) Train time/offline: **Augment** training data in context, to distill into parametric knowledge.

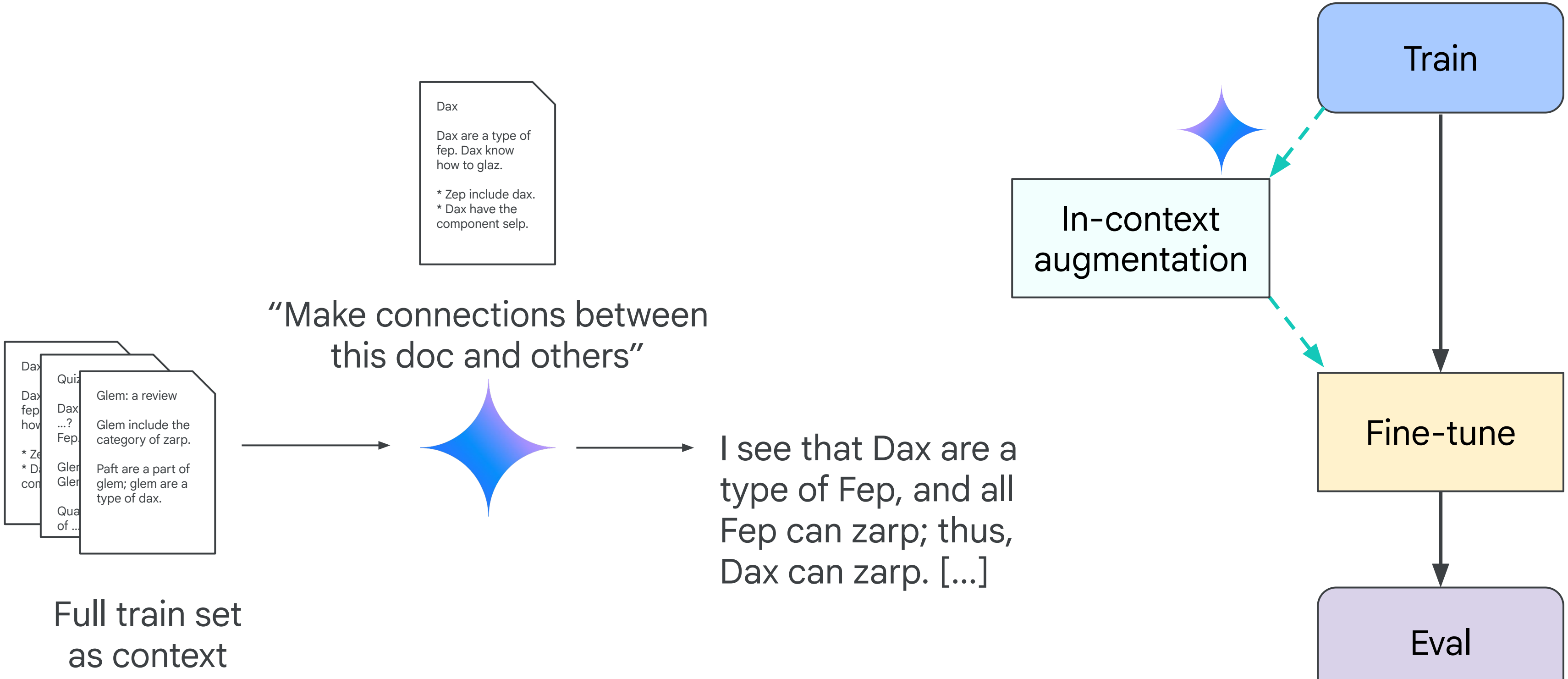
2) Test time/online: **Retrieve** memory at test time, to make it an in-context task. (Explicit retrieval from episodic memory, or implicit learned via RL.)

Approach #1: Can we improve fine-tuning generalization via augmentation?

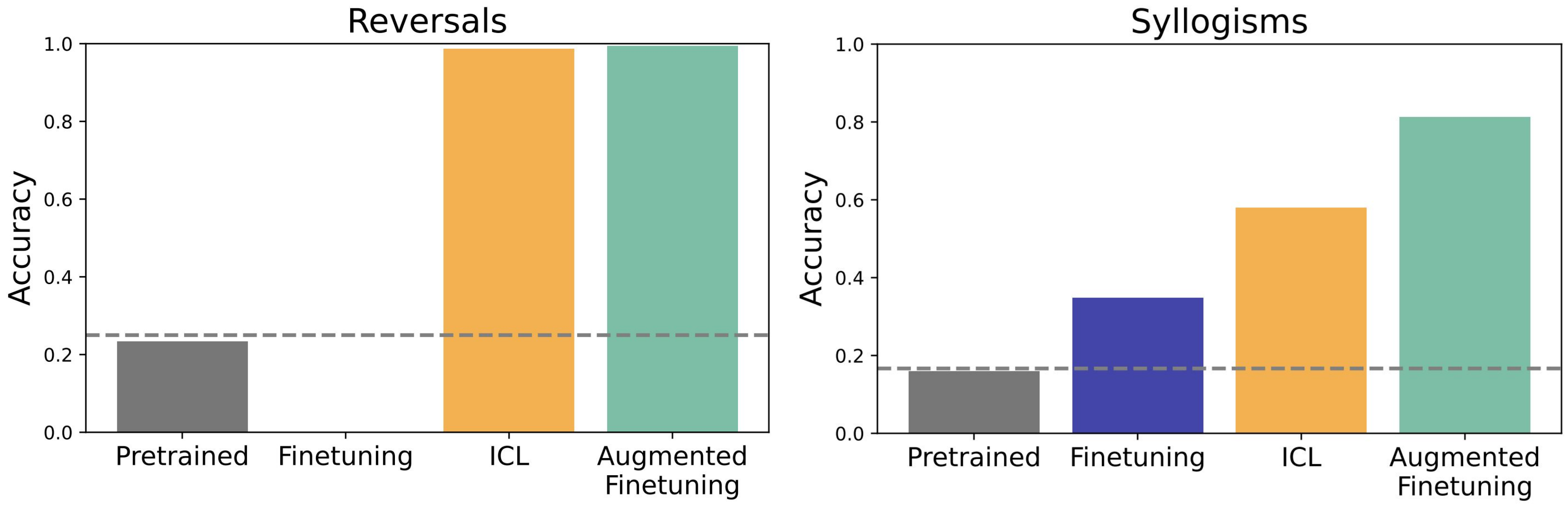
- Simple idea: if ICL generalizes better, just use in-context inferences to augment the original training data.
- Prompt model (with dataset in context) to generate reasoning traces elaborating on the training data, filling in reversals and missing links between documents.
- Add these new conclusions to the original dataset, and then finetune on this augmented dataset.



Approach #1: Can we improve fine-tuning generalization via augmentation?

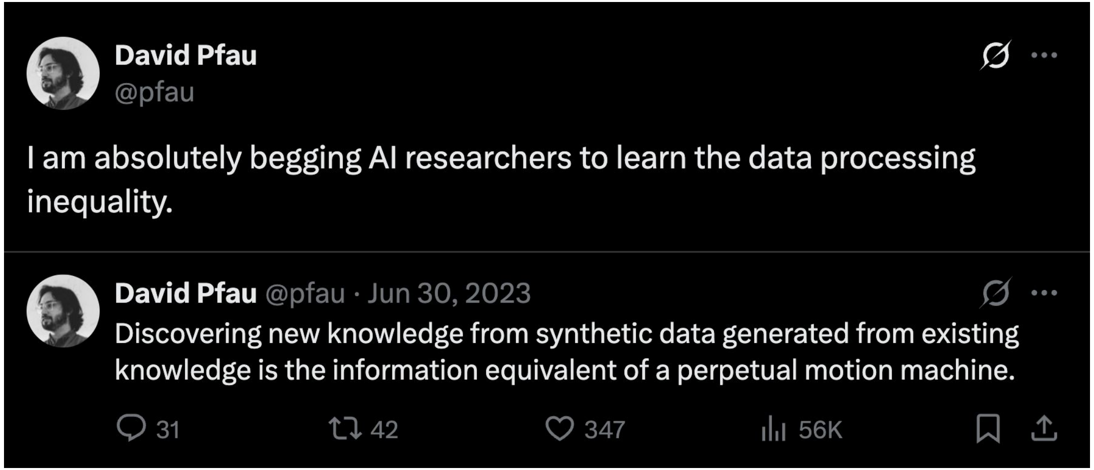


Augmented fine-tuning generalizes as well as ICL, or even better!

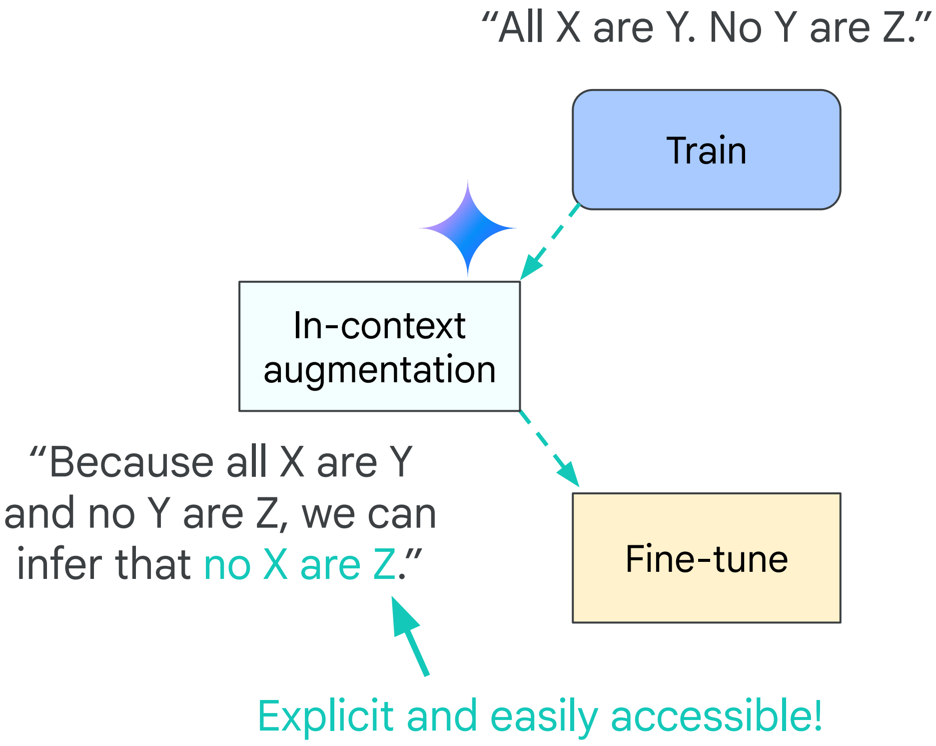


Using ICL to augment fine-tuning data can improve generalization.

How can we get more information from nothing?
Isn’t this theoretically impossible?????



Augmenting is not “creating” new information; instead, it’s making explicit & readily accessible what is already latent in the dataset



Trends in Cognitive Sciences
CellPress
Review

Learning by thinking in natural and artificial minds

Tania Lombrozo [1,*]

Published as a conference paper at ICLR 2020

A THEORY OF USABLE INFORMATION UNDER COMPUTATIONAL CONSTRAINTS

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The problem: having to imagine **any** way that current information could possibly be used in the future and then augment for it is intractable

Psychological Review
1995, Vol. 102, No. 3, 419–457

0033-295X/95/$3.00

Why There Are Complementary Learning Systems in the Hippocampus and Neocortex: Insights From the Successes and Failures of Connectionist Models of Learning and Memory

James L. McClelland
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University of Arizona

Randall C. O'Reilly
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and the Center for the Neural Basis of Cognition

Damage to the hippocampal system disrupts recent memory but leaves remote memory intact. The account presented here suggests that memories are first stored via synaptic changes in the hippocampal system, that these changes support reinstatement of recent memories in the neocortex, that neocortical synapses change a little on each reinstatement, and that remote memory is based on accumulated neocortical changes. Models that learn via changes to connections help explain this organization. These models discover the structure in ensembles of items if learning of each item is gradual and interleaved with learning about other items. This suggests that the neocortex learns slowly to discover the structure in ensembles of experiences. The hippocampal system permits rapid learning of new items without disrupting this structure, and reinstatement of new memories interleaves them with others to integrate them into structured neocortical memory systems.

One of the most striking neuropsychological phenomena ever reported is the dramatic amnesia produced by bilateral lesions to the hippocampus and related temporal lobe structures (Scoville & Milner, 1957). A crucial aspect of this phenomenon is temporally graded retrograde amnesia. Considerable evidence now supports the conclusion that the influence of the hippocampal system on the ability to exploit information derived from past experience in a wide range of tasks is temporally circumscribed: Performance is impaired if the hippocampal system is damaged before or within a window of time after the initial experience; however, if the hippocampal system is left intact both during the experience and for a period of time thereafter, subsequent damage may have little or no impact on performance.

This change in dependence on the hippocampal system over time appears to be a slow, gradual process. This gradual change has often been called consolidation, but the term really only labels the phenomenon. In this article, we focus on consolidation and consider what produces it and why it occurs. We ask, Is the phenomenon a reflection of an arbitrary property of the nervous system, or does it reflect some crucial aspect of the mechanisms of learning and memory? Is the fact that consolidation can take quite a long time—up to 15 years or more in some cases—just an arbitrary parameter, or does it reflect an important design principle?

We begin with a brief overview of the neuropsychology of memory, emphasizing the temporally circumscribed role of the hippocampal system, and elaborate one possible account of the functional organization of memory that is broadly consistent with the neuropsychological evidence, as well as aspects of the underlying anatomy and physiology. We then describe results from connectionist modeling research that suggest reasons for this organization and for the phenomenon of gradual consolidation. From the insights gained through the consideration of these models, we develop illustrative simulation models of the phenomenon of temporally graded retrograde amnesia. These

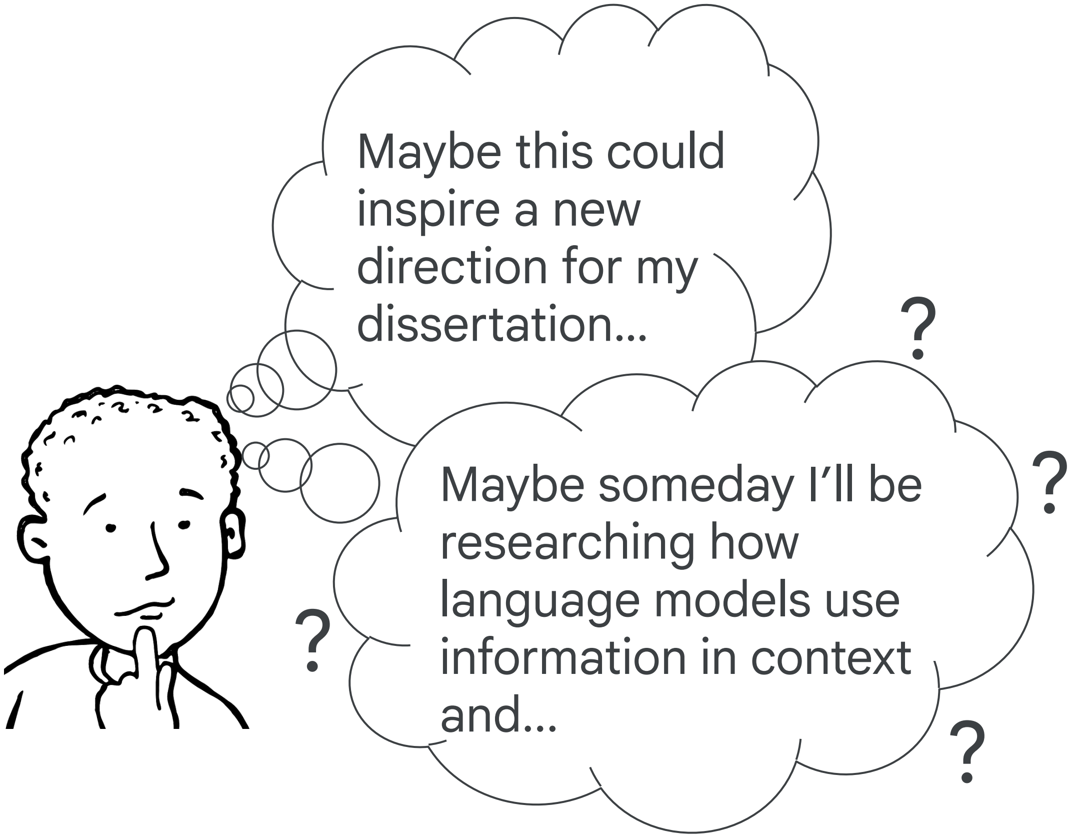
James L. McClelland and Randall C. O'Reilly, Department of Psychology, Carnegie Mellon University, and the Center for the Neural Basis of Cognition; Bruce L. McNaughton, Departments of Psychology and Physiology and Arizona Research Laboratories, Division of Neural Systems, Memory and Aging, University of Arizona.

The work reported herein was supported by National Institute of Mental Health Grants MH00385, MH47566, and MH46823; by a collaborative grant from the Human Frontiers Program; by a Center Grant from the McDonnell-Pew Program in Cognitive Neuroscience; and by a fellowship from the Office of Naval Research.

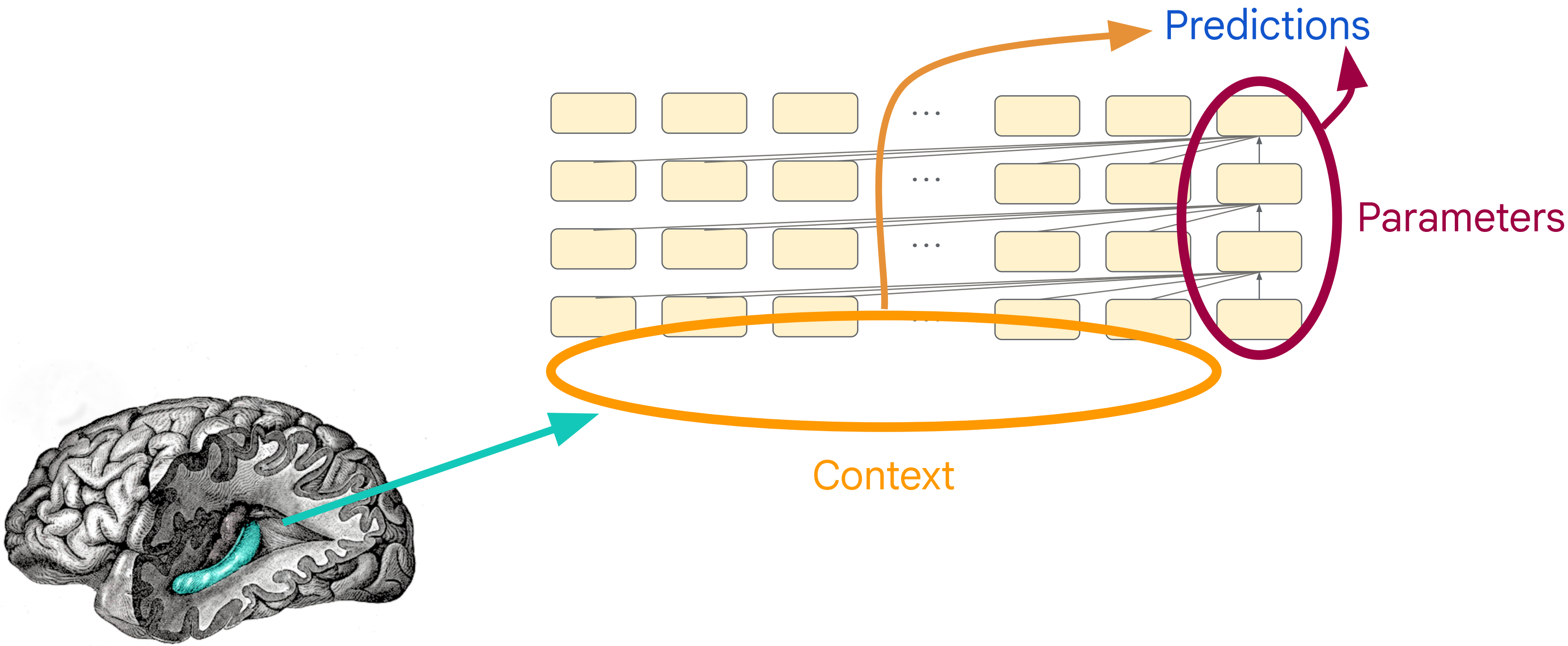
We thank Carol Barnes, Sue Becker, Marlene Behrmann, Loren Frank, Mark Gluck, Robert French, Patrick Lynn, Mort Mishkin, Catherine Myers, David Plaut, William Skaggs, Craig Stark, Robert Sutherland, Larry Squire, and Matthew Wilson for useful discussions or comments on earlier versions of this article. Lynn Nadel's contributions to the initial formulation of some of these ideas and to several revisions of the article were extensive and are gratefully acknowledged.

The computational arguments developed here were sketched in a Society for Neurosciences abstract (McClelland, McNaughton, O'Reilly, & Nadel, 1992) and an invited review of several projects involving James L. McClelland that referred to the present article as a work in progress (McClelland, 1994b). Preliminary versions of the simulations of retrograde amnesia in rats and monkeys were previously presented in a Psychonomic Society abstract (McClelland, McNaughton, & O'Reilly, 1993).

Correspondence concerning this article should be addressed to James L. McClelland, Department of Psychology, Carnegie Mellon University, Pittsburgh, Pennsylvania 15213. Electronic mail may be sent via Internet to mcclelland+@cmu.edu.

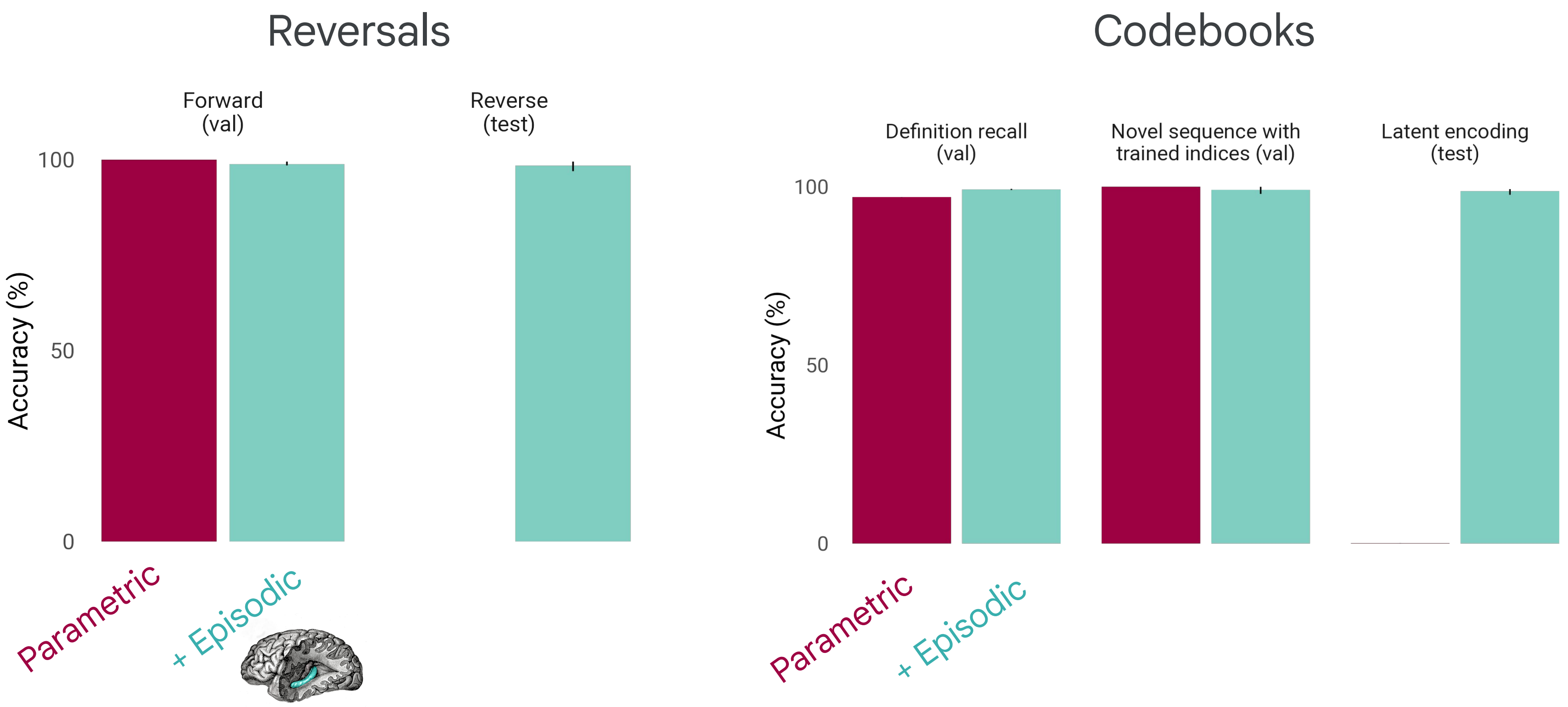


Approach #2: What if episodic retrieval can **put** the information that's needed in context, **just** when it's needed?



*Important disclaimer: we use an oracle episodic memory here to illustrate the point. It has perfect recall but imperfect precision (recalls at least one relevant memory + several irrelevant).

Episodic retrieval unlocks flexible generalization from latent information

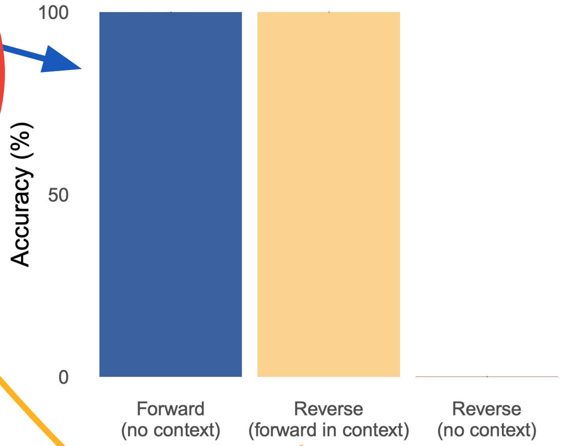


Episodic retrieval can bring learning experiences into context, where they can support flexible generalization.

Could models learn to do this via RL, without needing explicit retrieval (or augmentation)?

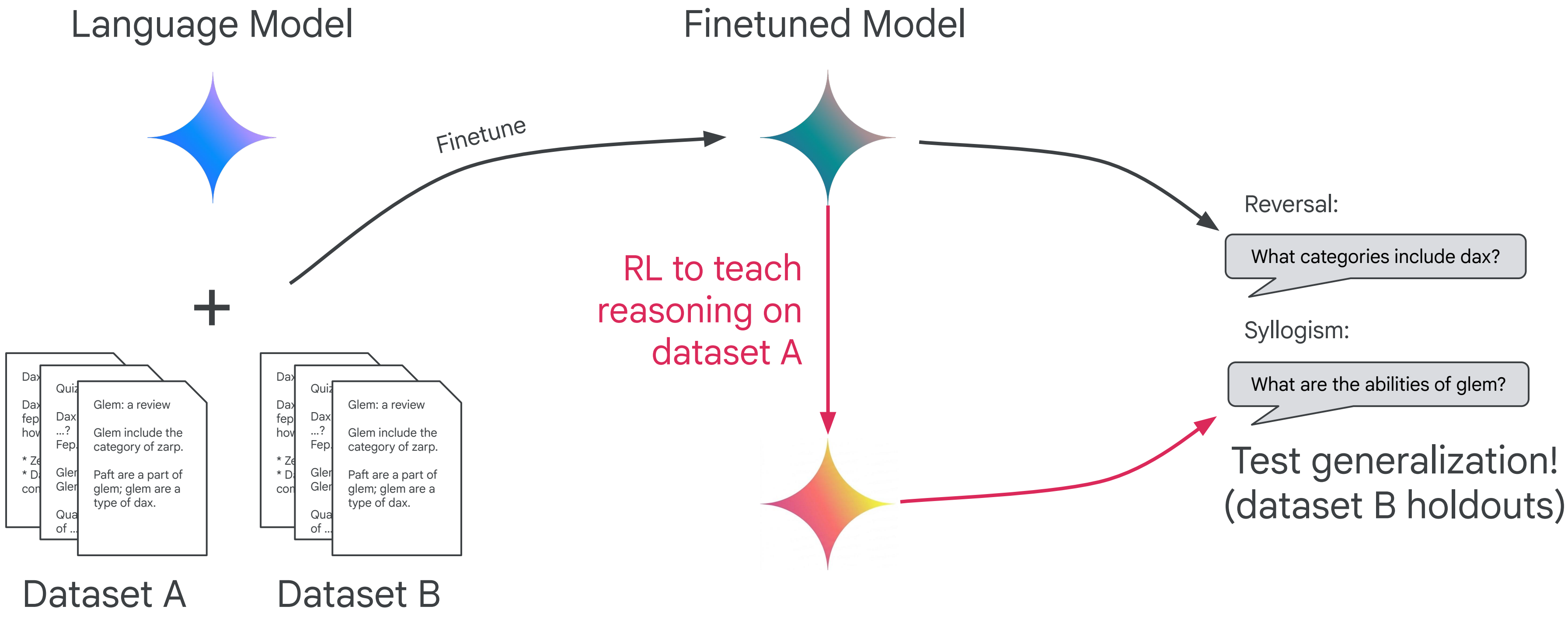
Trained relations:
forward relationship

ICL:
Forward + reverse

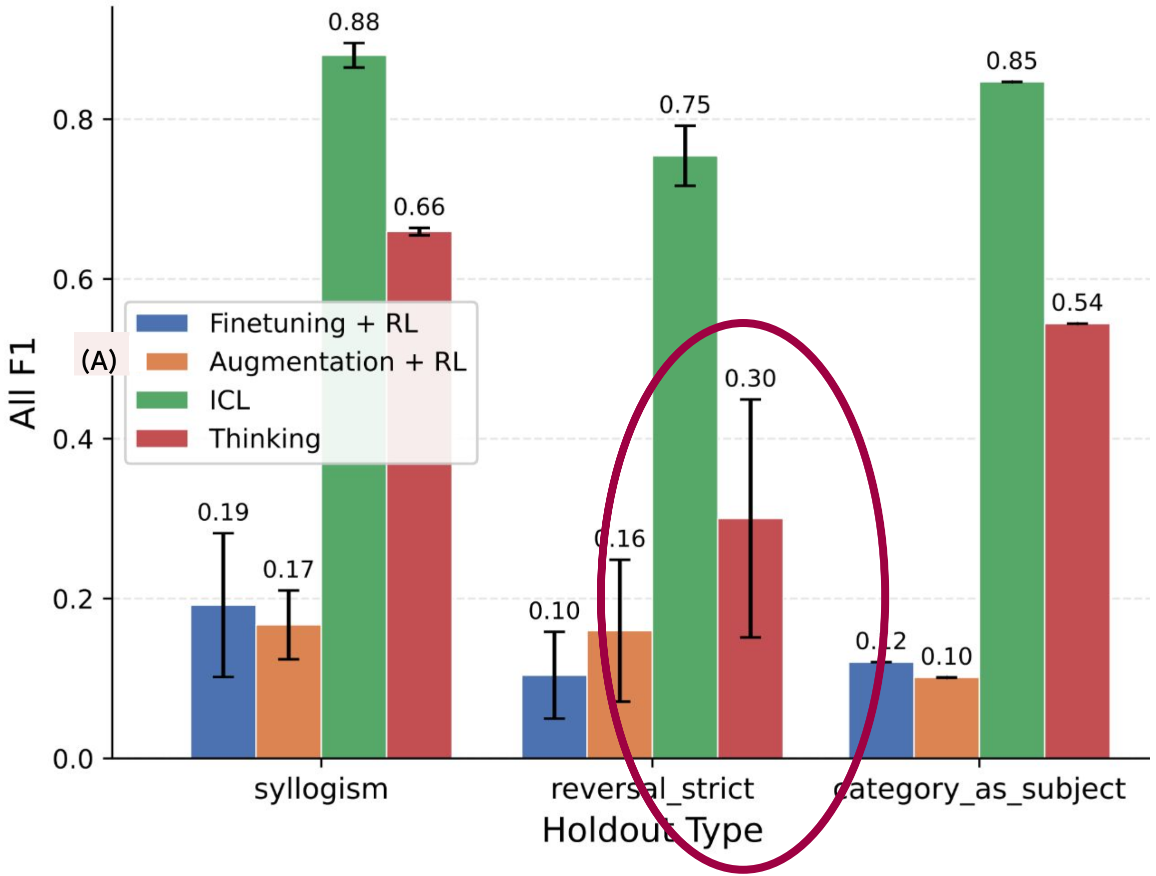


- Model already knows the pieces, **including** the information we retrieved.
- It just needs to learn to **regenerate** that necessary information in its CoT to bring it back into context if there is no retrieval; can it learn this via RL?

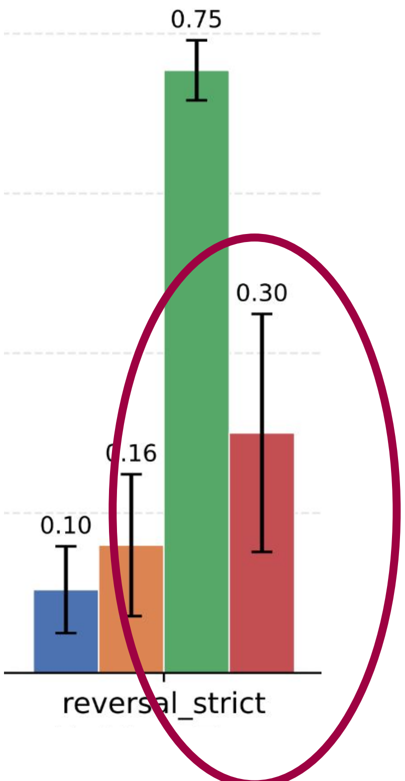
Instill multiple datasets of new knowledge (with holdouts) into model; use RL on one to try to improve generalization to the other



RL generalizes to dataset B holdouts! (augmenting **only** dataset A doesn't)



… but some problems are still hard. RL can only fix a true reversal curse via exhaustive enumeration — not very practical.



<thought>

Okay, the question asks for the name of a person based on a description. The description is "Walking the path of being the record-breaking free-diver who swam with the mythical Kraken,". So I need to find the name of the person who matches that description.

I remember reading about Cassidy Hammond. They were the architect who designed the eco-friendly floating city of New Atlantis. That doesn't fit the description of a record-breaking free-diver.

I remember Tessa Montgomery. They were the world-champion hoverboard racer. Not a free-diver.

Hold on, I recall Maxwell Alderwood. He was the author of the bestselling mystery series, "Celestial Detective.". That's a writer, not a diver.

How about Tyler Oakridge? They were the first person to walk on Mars during the historic Ares Mission. That's an astronaut, not a diver.

⋮ (several pages omitted)

Dominic Mullins! I remember reading about him. He was the record-breaking free-diver who swam with the mythical Kraken. Yes, that's it!

Okay, it was definitely Dominic Mullins.

</thought>

Dominic Mullins

Models can learn via RL to regenerate information they need in context at test time; this ameliorates some problems, but not all.

Paths to better generalization by leveraging in-context reasoning:

1) Train time/offline: **Augment** training data in context, to distill into parametric knowledge.

2) Test time/online: **Retrieve** memory at test time, to make it an in-context task. (Explicit retrieval from episodic memory, or implicit learned via RL.)

Varying pros and cons:

Train-time augmentation:

- Achieves performance as good or better than ICL! (On questions about augmented data)
- Requires a lot of extra compute at training time.
- Efficient at test time!

Test-time retrieval:

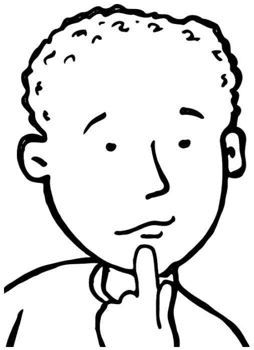
- Achieves performance as good as ICL!
- Free at training time!
- Requires extra inference compute at test time.
- We don't give an answer for how to do retrieval well in a general way.

Test-time thinking via RL:

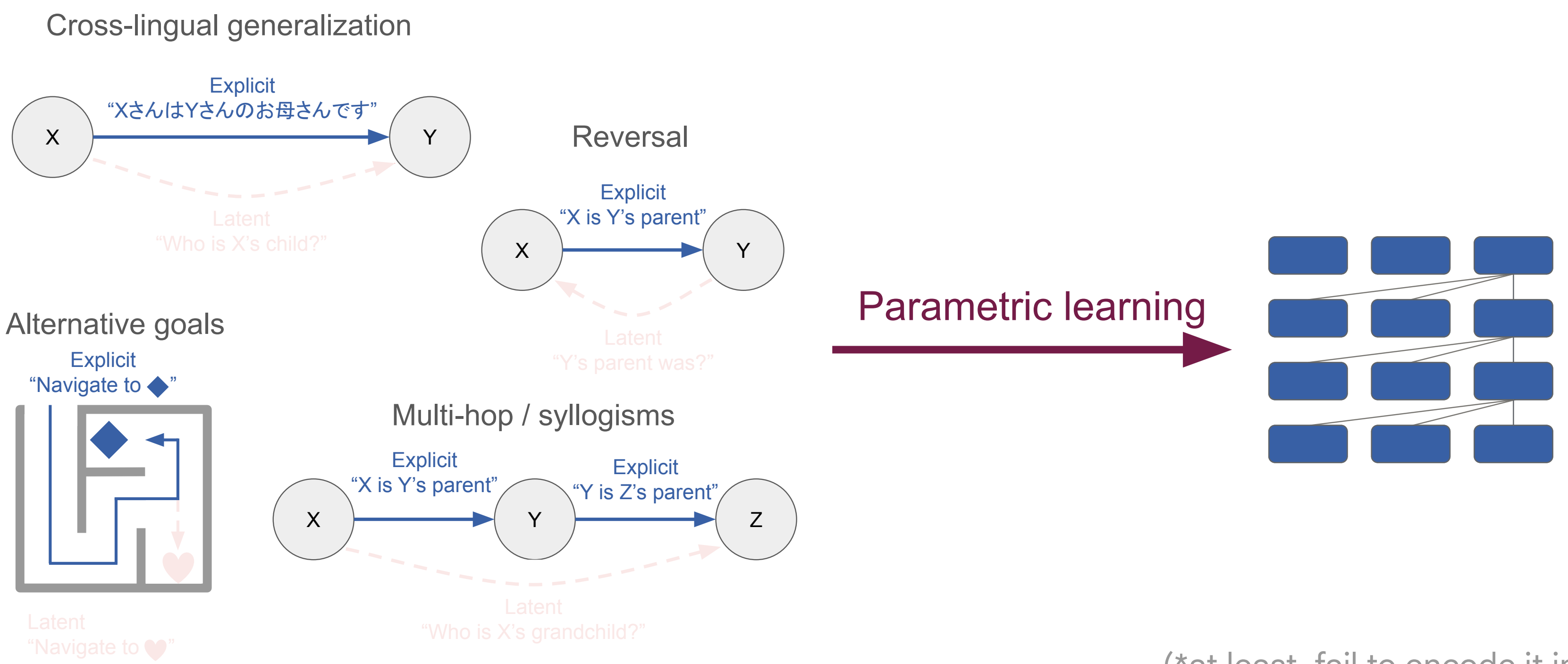
- Good performance on many types of structures!
- Much worse than ICL on others, e.g. reversal.
- Generalizes beyond the augmented / explicitly retrieved distribution.
- Requires extra inference compute at training and test time.

03

The bigger picture, and natural intelligence

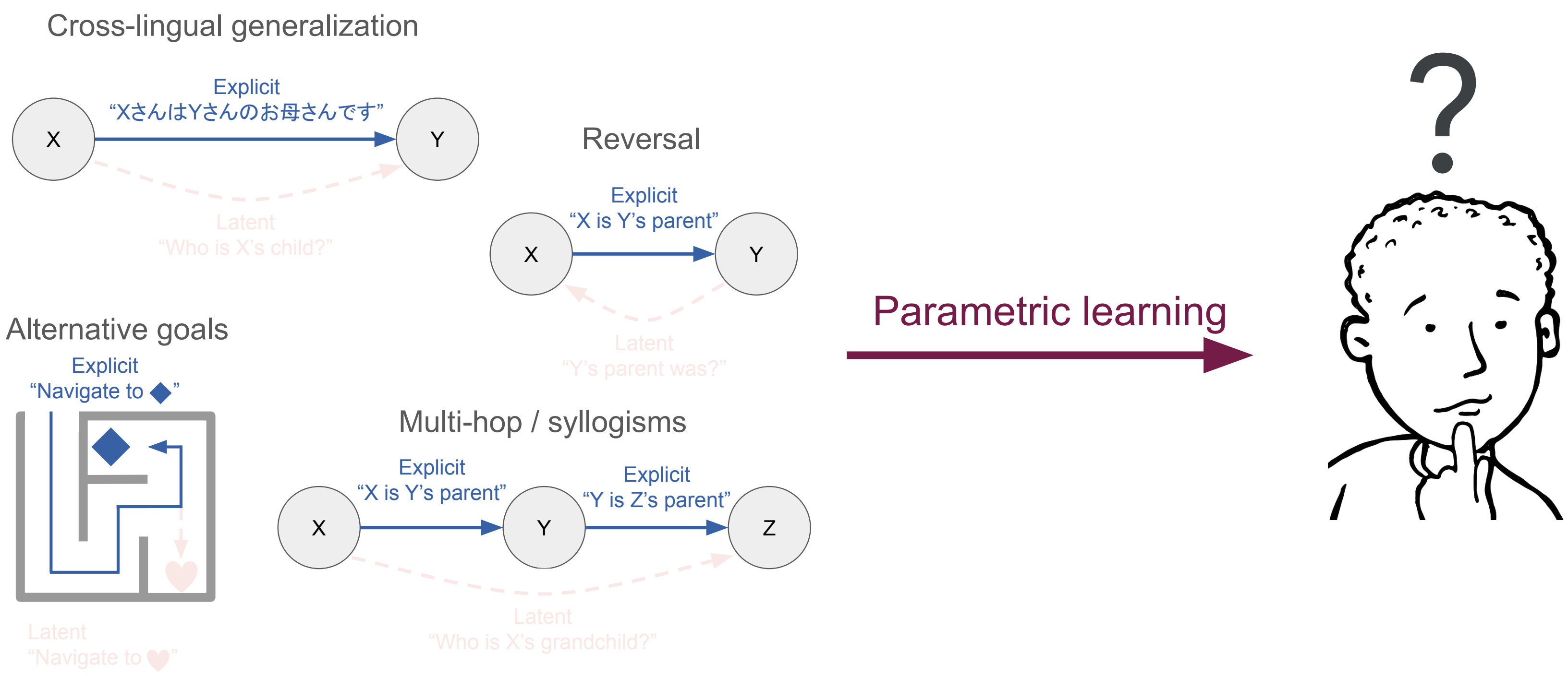


Training data often *latently* conveys more than its explicit content. LM parameters may fail* to effectively encode this latent information.



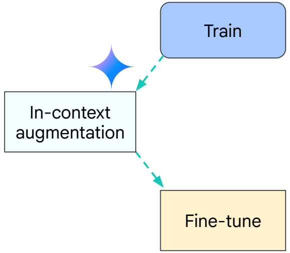
(*at least, fail to encode it in a way that’s reliably usable at test time)

Does natural intelligence face the same challenges?



If so, might there be evidence that natural intelligence also uses these paths to bridge the generalization gap?

Train time/offline: **Augment plausible**

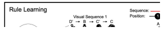


Test time/online: **Retrieve relevant**

Cell

Human Replay Spontaneously Reorganizes Experience

Graphical Abstract



Offline replay supports planning in human reinforcement learning

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Medial Temporal Lobe Amnesia: Gradual Acquisition of Factual Information by Nondeclarative Memory

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Preemptive Solving of Future Problems: Multitask Preplay in Humans and Machines

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Neuron

Article

Hippocampal Replay Is Not a Simple Function of Experience

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Hippocampal representations of alternative possibilities are flexibly generated to meet cognitive demands

Alison E. Comrie[1], Emily J. Monroe[2], Ari E. Kahn[3], Eric L. Denovellis[4], Abhilasha Joshi[4], Jennifer A. Guidera[5], Timothy A. Krausz[1], Joshua D. Berke[6,7], Nathaniel D. Daw[3,8], Loren M. Frank[2,4,6,9]

The roles of online and offline replay in planning

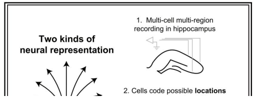
Eran Eldar[1,2,3]*, Gaëlle Lièvre[2,3], Peter Dayan[4,5]†, Raymond J Dolan[2,3]†

Cell

Article

Constant Sub-second Cycling between Representations of Possible Futures in the Hippocampus

Graphical Abstract



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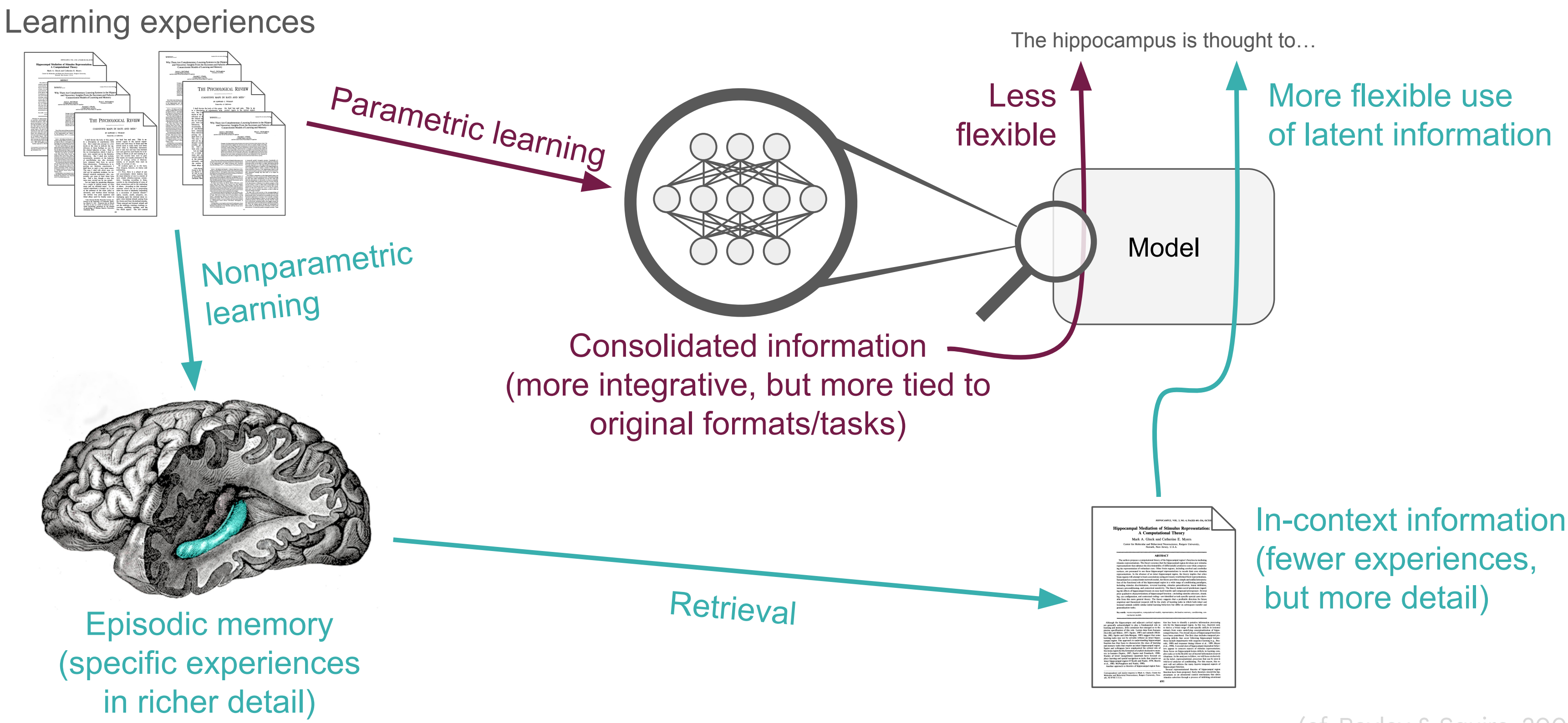
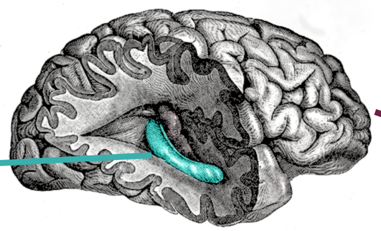
(*apologies to the neuroscientists for my gross characterization of these complex works)

A perspective on why cortical and hippocampal learning are complementary



(cf. Bayley & Squire, 2002, etc.)

What computational ingredients are needed for generalization?

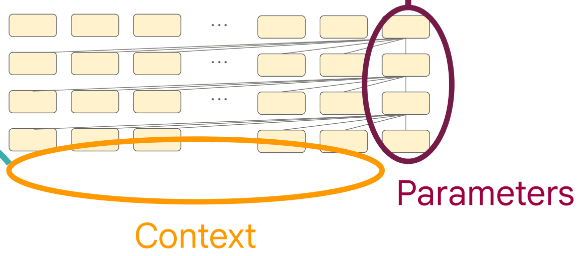


Episodic memory:

- Experiences in rich detail, allowing access to information we didn't know would be important later.

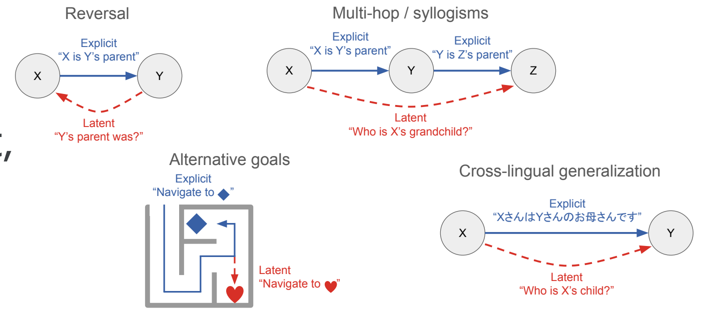
Consolidated information:

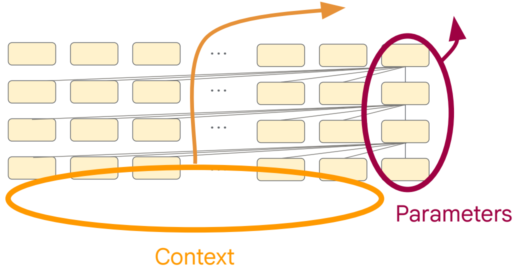
- Cached facts and insights within or across experiences.
- Including procedures for reasoning flexibly about information in context.

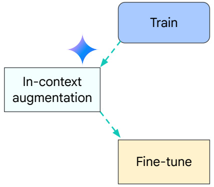


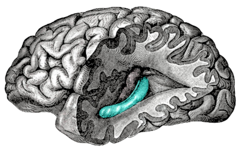
Summary:

- Training experiences contain both explicit content, and latent information that might be useful later.
- Language models may not reason over latent information effectively after parametric learning, but can use it flexibly in context.
- We showed several routes to bridge this gap:
  - Offline: augment training data.
  - Online: retrieve relevant experiences (explicitly via episodic memory, or implicitly, learned via RL)
- These might relate to offline/online hippocampal replay, and illustrate a different way that hippocampus & neocortex are complementary.
- Though of course brains might also employ other solutions, e.g. better learning rules...









Thank you.

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